LM Physics of Climate - Problem Sheet 1

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*Questions marked with * are more challenging and closer to the style of question (or parts of a question) you can expect in the exam. Unmarked questions are shorter and designed to quickly check your understanding of a particular concept.*

Problem 1.1 - Earth's surface temperature

In Section 1.2, we used a simple climate model to derive surface temperature as a function of equilibrium temperature, T_{eq}, and the efficiency of the atmosphere's emission/absorption relative to a blackbody, ε_{IR} (Equation 1.5). What value does ε_{IR} need to take to correctly give the Earth's surface temperature?

Problem 1.2* - Planetary interior flux

A planet receives energy from its host star, but also emits its own thermal energy due to processes such as radioactive decay in its interior. At the surface of the planet, the contribution to the energy flux from the planetary interior is σT_{int}^4.

(i) Assuming that the planet is a distance a from its host star, which has radius $R_\star$ and temperature $T_\star$, what is the temperature of the planet? You can assume that the planet and star act as a blackbody radiators.

(ii) For the Earth, $T_{int} \approx 10$ K. Is it reasonable to ignore the internal heat when calculating the Earth's temperature?

Problem 1.3 - Scale height

The Earth's mean molecular weight, $\mu \approx 28.97$ u. Would the vertical extent of the atmosphere be thicker or thinner, and by what factor, if the atmosphere were made of pure CO_2 ($\mu = 44$ u)? The atmosphere of Venus is dominated by CO_2, and yet near its surface it has a similar scale height (within a factor $\sim$2) to the Earth. Why?

Problem 1.4 - Assuming constant gravity

We often assume in this course that the gravitational acceleration is constant throughout the atmosphere. Demonstrate whether this is true by calculating the change in g between the surface and the top of the Earth's atmosphere, which is often defined to be at 100 km. You can assume that the mass of the atmosphere is negligible compared to the rest of the planet, as you will show in Problem 1.6.

Problem 1.5* - Pressure profile for a constant lapse rate

The temperature in an atmosphere has a constant lapse rate, $-\frac{\mathrm{d}T}{\mathrm{d}z} = \Gamma$. The gravitational acceleration, g, and mean molecular weight, μ, can be assumed to be constant with height.

(i) Assuming the atmosphere is in hydrostatic equilibrium, calculate pressure as a function of height, z. Give your answer in terms of the surface temperature, T_0, surface pressure, P_0, the lapse rate, Γ, g and μ.

(ii) Where no condensation occurs, the Earth has a 'dry' lapse rate of $\Gamma = 9.8$ Kkm^{-1}. Using this value, what are the temperature and pressure at the top of Mount Everest (8.8 km), assuming $\mu = 28.97$ u and $T_0 = 288$ K?

Problem 1.6* - Mass of the Earth's atmosphere

(i) Calculate the mass of the Earth's atmosphere, assuming it is hydrostatic and isothermal. Write your answer in terms of the planetary radius, surface pressure, and gravitational acceleration.

(ii) The average radius of the Earth is 6371 km, and the surface pressure is 1 bar= 10^5 Pa. What is the total mass of the atmosphere? Verify that it is negligible compared to the total mass of the Earth (as we assumed in Problem 1.4).

Problem 1.7* - Sound speed in an ideal gas

Sound waves in a gas can be assumed to be adiabatic.

(i) Show that the sound speed, $v = \sqrt{\frac{\mathrm{d}P}{\mathrm{d}\rho}}$, for an ideal gas can be written as:

$$v = \sqrt{\gamma g H_{sc}},$$

where γ is the adiabatic index of the gas, g the gravitational acceleration, and H_{sc} the scale height of the atmosphere in which the sound waves are propagating.

(ii) Mars and Venus both have CO_2-dominated atmospheres with $\gamma \approx 1.3$. What are the sound speeds at the base of each of their atmospheres?